In contrast, in RL we don’t know exactly what the right thing to do is at every step. We just need to know what the ultimate goal is and what things to avoid doing. How do you teach a dog a trick? You have to give it tasty treats. Similarly, as the name suggests, we train an RL algorithm by incentivizing it to accomplish some high-level goal and possibly disincentivize it from doing things we don’t want it to do. In the case of a self-driving car, the high-level goal might be “get to point B from starting point A without crashing.” If it accomplishes the task, we reward it, and if it crashes, we penalize it. We would do this all in a simulator, rather than out on the real roads, so we could let it repeatedly try and fail at the task until it learns and gets rewarded.

Tip

In natural language, “reward” always means something positive, whereas in reinforcement learning jargon, it is a numeric quantity to be optimized. Thus, a reward can be positive or negative. When it is positive, it maps onto the natural language usage of the term, but when it is a negative value, it maps onto the natural language word “penalty.”

The algorithm has a single objective—maximizing its reward—and in order to do this it must learn more elementary skills to achieve the main objective. We can also supply negative rewards when the algorithm chooses to do things we do not like, and since it is trying to maximize its reward, it will learn to avoid actions that lead to negative rewards. This is why it is called *reinforcement learning*: we either positively or negatively reinforce certain behaviors using reward signals (see figure 1.7). This is quite similar to how animals learn: they learn to do things that make them feel good or satisfied and to avoid things that cause pain.

Figure 1.7. In the RL framework, some kind of learning algorithm decides which actions to take for a control task (e.g., driving a robot vacuum), and the action results in a positive or negative reward, which will positively or negatively reinforce that action and hence train the learning algorithm.